

DS 302: Computing Systems for Data Processing

Course Code: DS 302

Course Name: Computing Systems for Data Processing

L-T-P-C: 2-0-2-3

Prerequisites: None

Intended for: UG

Distribution: Discipline core for B. Tech. Data Science and Engineering, Not allowed to B.Tech. CSE, Discipline elective for B. Tech. Electrical Engineering, Free elective for other B. Tech. disciplines.

Semester: Even/Odd

Approval: 29th BoA as Computing for Data Science (B.Tech DSE curriculum approved by the 20th Senate); re-approved in the 55th BoA for the DSE 2022 and earlier batches; credits recorded as 3 (corrected from 4) in the 65th BoA

Preamble

One of the important problems in data science is how to design computing systems that can deal with various sources of data, and can scale to huge volumes of data. This course is designed to give basic understanding of storage and computing systems at the logical and physical level of abstraction. Relevant topics from Operating Systems, Computer Organization and Databases are fused in this course to fulfil the above course objective. On completion of the course, students should be able to understand how data is represented in the computer, the different memory hierarchies and their performance, trade-offs between using processes and threads for applications, concepts of scheduling and different data models like Relational and NoSQL.

Course Modules

- **Basics of Computer Organization:** Data representation, machine code, computer arithmetic, code compilation, memory organization and management, memory and run time performance optimization (8 lectures)
- **Introduction to Operating Systems:** Responsibilities of OS; process & thread management: process model, states and its structure, process creation and termination, thread models and issues, User/Kernel level threads; inter-process communication, process synchronization, and process scheduling; file systems. (8 lectures)
- **Introduction to Databases:** Information modeling, ACID properties and Transactions, Trade-offs between Relational Databases and NoSQL, Data Manipulation Language SQL (8 lectures)
- **Scalable data processing using MapReduce:** Hadoop Distributed File System, MapReduce programming model (4 lectures)

Text Books

1. Stallings, William. *Computer Organization and Architecture*, Global Edition. Pearson Education Limited, 2015.
2. Silberschatz, Abraham, Greg Gagne, and Peter B. Galvin. *Operating system concepts*. Wiley, 2018.
3. Elmasri, Ramez, and Sham Navathe. *Fundamentals of database systems*. Pearson, 2017.



Reference Books

1. Forouzan, Behrouz, Catherine Coombs, and Sophia Chung Fegan. *Introduction to data communications and networking*. McGraw-Hill, Inc., 1997.
2. Ramakrishnan, Raghu, and Johannes Gehrke. *Database management systems*. McGraw Hill, 2000.
3. Carl Hamacher, V., Zvonko G. Vranesic, and Safwat G. Zaky. *Computer organization*. 2004.